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# Slave trade and the spread of Islam in Africa*

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**Abstract**

We investigate an often overlooked legacy of the Atlantic slave trade: its role in the spread of Islam in West Africa. Building on the historical literature, we examine whether the protection offered by Islamic rulers against enslavement served as an incentive for conversion to Islam. Using a theoretical model, we analyze individuals' motivations to embrace Islam in response to the threat of slavery. The empirical analysis reveals a negative relationship between the intensity of the Atlantic slave trade and Islamic prevalence, showing that Muslim regions exported fewer slaves. Additionally, areas under Islamic rule, where individuals had de facto protection from enslavement, saw higher conversion rates. These findings are consistent across both historical data on Islamic prevalence after the end of the slave trade (in the early 20th century) and contemporary survey data on religious affiliation. Our results highlight how the Atlantic slave trade not only shaped Africa's socio-economic development but also significantly influenced the religious landscape, with lasting effects on Islam's diffusion.

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# 1 Introduction

The transatlantic slave trade arguably stands as one of the largest intercontinental economic enterprises of the early modern world (Zeuske, 2023). Its vast scale marked a profound rupture in African history (Lovejoy, 1989), leaving an enduring impact on African societies. Recent empirical studies suggest that Africa's slave trades not only contribute to its current underdevelopment (Nunn, 2008) partly by fostering a culture of mistrust (Nunn and Wantchekon, 2011), but are also linked to reduced access to formal trade credit needed by modern firms, hindering capital investment in smaller enterprises (Pierce and Snyder, 2018). The slave trades led to prolonged periods of abnormal sex ratios, which influenced polygyny rates across Africa (Dalton and Leung, 2014) and affected women's labor force participation, fertility rates, and involvement in household decision-making (Teso, 2019).

An important but less known legacy of Atlantic slavery is the way it shaped people's religious affiliation, specifically how it acted as a catalyst to the continents' Islamization.[1] Islam acted as a protective force against slavery, particularly in West Africa. Muslim commercial activity and political shifts, particularly those associated with *jihad*, curtailed slave exports across the Atlantic. Regions with significant Muslim influence or merchant presence experienced comparatively lower levels of slave exports to the Americas than might have otherwise been expected (Lovejoy, 2002). Simultaneously, the protection afforded to Muslims under Islamic rule provided an incentive for conversion. This paper examines these two hypotheses: (i) fewer slaves were captured and exported from Islamic regions to the New World, and (ii) protection against enslavement enhanced Islam's appeal, leading to higher conversion rates. We provide empirical evidence supporting both arguments, demonstrating that slavery contributed to the spread of Islam in Africa, an effect that persists to this day.

We develop a simple two-period theoretical model describing an individual's incentives to convert to a different religion when their community is under threat of an attack by slave traders. We then assess the model's implications through empirical analysis. After mapping the strength of pre-existing Islamic beliefs before the onset of the slave trade, we provide comprehensive evidence of Islam's expansion during the 16th to 19th centuries, using a rich array of primary and secondary sources. Our findings reveal a negative relationship between the Atlantic slave trade and Islamic prevalence, confirming that Muslim areas exported fewer slaves to the Atlantic econ-

[1] Existing work has examined the influence of Christian missionary evangelisation on Africans' religious beliefs of (Nunn, 2010).

omy, and a positive one between de facto protection from enslavement (proxied by rulers' religious affiliation) and the spread of Islam, suggesting that people embraced Islam as a safeguard against slavery. These results hold both for historical mesures of Islamic prevalence (in the first half of the 20th century) measured at the ethnic homeland level, and for contemporary years, measured at the individual level, using survey data on religious affiliation, thus highlighting another persistence effect of Atlantic slavery.

## 2 Background

*It is incumbent upon Muslims to free enslaved fellow Muslims*
Muhammad Bello, 2nd Caliph of Sokoto[2]

The Islamization of West Africa progressed slowly compared to other regions that were closer to the Arabian Peninsula. It is not until the eleventh century that some rulers of West Africa began to convert (starting with the first king of Gao in 1,000 CE)[3] and that in East Africa Islam emerged as a unifying force forming a distinct Swahili identity (Trimingham, 1964). Until the 13th century, the presence of Islam was limited to segregated Muslim communities linked to the trans-Saharan trade, namely a Muslim merchant-scholar class residing in non-Muslim kingdoms. Over the following centuries, African rulers began to adopt Islam while ruling over populations with diverse faiths and cultures. Rulers' conversion facilitated the spread of Islam among Africa's population

Trade conducted along pre-Islamic commercial routes has been identified as one of the main drivers behind the spread of Islam in Africa (Michalopoulos et al., 2018). While trade was instrumental in the transmission of the Islamic faith, its delayed spread suggests that other factors likely contributed to its uptake by large segments of the African population.

The process of Africa's Islamization gained momentum in the seventeenth century, and was shaped by two interconnected forces: the development of navigation technologies that connected the African coast with Europe (Salvaing, 2020) and the onset of the Atlantic slave trade (Lovejoy, 2002); both phenomena created a struggle between Christianity and a form of Islam that became increasingly political. During this period a series of armed *jihad* enabled the expansion of Islam towards the continents' south. Such jihad revolutions can be seen as a direct response to the Atlantic slave trade (Lovejoy, 2009) to the point that have been labelled as "anti-slavery

[2]Cited in Naqar (1972, p.55)
[3]The empire of Gao was an important predecessor of the Songhai Empire.

movements", an argument first introduced by Humphrey Fisher in 1989 (Lofkrantz, 2023).[4]

The newly born Islamic governments established through revolutions, such as the Sokoto Caliphate (1804-1903), the Caliphate of Hamdallahi (1820-1862) and the Umarian States (1850s-1890s) did not outlaw slavery, but regulated enslavement and the trade of slaves according to the interpretation of the Maliki school of law, which safeguarded people considered to be freeborn Muslims (Lofkrantz, 2023).[5] Atlantic slavery was such a prominent topic for West African governments that, for instance, in the Sokoto Caliphate it was debated not less than it was in Britain in the nineteenth century, with the focus of the debate revolving around religious identity, specifically on the emancipation and protection of Muslims (Lofkrantz, 2011).

The protection of the rights of freeborn Muslims was considered a duty for leaders of Muslim rulers, and those who did not do so were labelled as "unbelievers".[6] In fact, the West African jihad adopted a series of measures to protect Muslims, as they: (a) made it illegal to enslave freeborn Muslims and to sell them towards the Atlantic slave trade; this was specifically motivated by their desire to encourage the conversion of non-Muslims to Islam; (b) instituted export checks to stop the export of illegally enslaved Muslims; (c) supported the ransoming of captive freeborn Muslims before they were enslaved. All these measures made Islam an attractive option for non-Muslims and plausibly increased the probability of conversion. A clear manifestation of this was the use of facial and body scarification to signal a person's faith: Muslims marked their children with patterns distinguishing them from those used by groups targeted for slavery (Lovejoy, 2009).

While well into the seventeenth century the demand for African slaves was predominantly met by African slave markets along the coast, by the late seventeenth/early eighteenth century also the continent's interior became a considerable slave supplier. Given that most African slaves were acquired as war captives, the advent of transcontinental slave trade generated further incentives to engage in warfare, shaped by the prospect of acquiring valuable slaves for export overseas (Zeuske, 2023).[7] The armed *jihad* can be seen within this context, as an effort to building

[4]To be clear, the term anti-slavery does not imply that the jihad revolutions were against slavery *per se*, as demonstrated by their continuous reliance on slaves throughout their government, but rather against the sale of slaves to non-Muslims and against the enslavement of freeborn Muslims (Lovejoy, 2009).

[5]West African jihads also led to the establishment of the Imamates of Fuuta Bundu (1698-1699), Fuuta Jaalon (1726-1727, and Fuuta Toro (1769-1776), located in today's Guinea and Senegal.

[6]The work of Islamic jurists Muhammad al-Maghili and Ahmad Baba describes in detail the prohibition on Muslims' enslavement and rulers' obligations to protected their Muslims subjects (Lofkrantz, 2011).

[7]The increase in the use of violence for the purpose of acquiring slaves manifested not only

states capable of countering the export of slaves to the Christian realm (Salvaing, 2020).[8]

# 3 Model

To describe the incentives that locals had to convert to a different religion, we consider a simple two-period theoretical model of a community that is under threat of an attack by slave traders. In the first period, the members of the community play a *conversion game*, where they simultaneously choose whether to convert to a different religion. Once all conversions have been finalized, the second period ensues where an attack from slave traders occurs with some probability. If an attack does occur, a fraction of the population is taken as slave.

Each community has a continuum of members of measure 1 and a ruler. At the outset of period 1, each member has a religion $r = m, \overline{m}$ (Muslim and non-Muslim respectively), which may or may not be equal to the ruler's religion $r^* = m, \overline{m}$. The share of Muslims in the community is denoted by $\mu \in [0, 1]$. Rulers are assumed to have a weak preference for protecting those who share the same religion in the event of an attack. Specifically, we assume that $\overline{\eta}_{r^*} > \underline{\eta}_{r^*}$, where $\underline{\eta}_{r^*}$ and $\overline{\eta}_{r^*}$ represent the probability of being taken as slave, conditional on an attack, if one shares and does not share the same religion $r^* = m, \overline{m}$ as the ruler respectively. It is convenient to define $\eta_{r^*} \equiv \overline{\eta}_{r^*} - \underline{\eta}_{r^*} > 0$ as a measure of the preferential treatment received by those who have the same religion as the ruler.

By converting to a different religion, each person incurs in some dis-utility $c$. This dis-utility represents a measure of *religious fervour* - the more important religion is for the person relative to the pain of being taken as a slave, the higher should be the cost of converting. We assume that each conversion dis-utility $c$ is independently drawn from some continuous distribution $F$ with full support on the interval $[0, 1]$. The dis-utility of being taken as slave is normalized to 1.

In this game, a strategy for each person is a function $\sigma : [0, 1] \times \{m, \overline{m}\} \to \{0, 1\}$,

in the form of conflict, but also via a higher rate of raiding, kidnappings and as forms of judicial punishment.

[8]It is important to highlight that the states born out of these holy wars did not abolish slavery, but managed to divert slave trade from an export oriented activity to one used for the domestic economy. Given that domestic demand was lower than international demand, it likely reduced slavery overall. It is indeed argued that the political transformations associated with *jihad*, have curtailed the export of slaves towards the Atlantic and explains the comparatively low levels of exports to the Americas from Muslim areas (Lovejoy, 2002). Furthermore, the relatively high incidence of emancipation and the fact that the children of concubines were considered to be free-born tended to limit the growth of the enslaved population (Lovejoy, 2009).

where $\sigma(c, r)$ represents the probability of conversion of a person with religion $r$ and religious fervour $c$. A symmetric Bayes-Nash equilibrium strategy $\sigma^*$ is such that $\sigma^*(c, r)$ maximizes the expected payoff of a person with religion $r$ and religious fervour $c$ when every other member of the community follows strategy $\sigma^*$.

Once conversion decisions are made, the new share of Muslims in the community is given

$$\widehat{\mu} \equiv \int_{c\in[0,1]} (\mu(1 - \sigma(c, m)) + (1 - \mu)\sigma(c, \overline{m}))dF(c).$$

In the second period, slave traders attack with probability $p(\widehat{\mu}) \in (0, 1)$.

**Proposition 1.** *Every symmetric Bayes-Nash equilibrium $\sigma^*$ is such that:*

*i) $\sigma^*(c, r^*) = 0$ for all $c \in (0, 1]$.*

*ii) If $r \neq r^*$, then*

$$\sigma^*(c, r) = \begin{cases} 1 \, \forall c < \overline{c}, \\ 0 \, \forall c > \overline{c}, \end{cases}$$

*for some $\overline{c} \in (0, 1)$.*

*Proof.* Let $\sigma^*$ be such that everyone playing strategy $\sigma^*$ is a Bayes-Nash equilibrium. Let

$$\widehat{\mu}^* \equiv \int_{c\in[0,1]} (\mu(1 - \sigma^*(c, m)) + (1 - \mu)\sigma^*(c, \overline{m}))dF(c)$$

denote the end-of-the-period share of Muslims in the community, so that, in the second period, an attack happens with probability $p(\widehat{\mu}^*)$.

Because there is a continuum of community members, each individual person cannot change the likelihood of an attack by converting. As a result, if $r = r^*$, the payoff for converting is always lower than the payoff of not converting, i.e.,

$$-p(\widehat{\mu}^*)\overline{\eta}_{r^*} - c < -p(\widehat{\mu}^*)\underline{\eta}_{r^*}.$$

However, if $r \neq r^*$, the community member prefer to convert if and only if

$$-p(\widehat{\mu}^*)\underline{\eta}_{r^*} - c \geq -p(\widehat{\mu}^*)\overline{\eta}_{r^*} \Rightarrow c \leq p(\widehat{\mu}^*)\eta_{r^*}.$$

As a result, it follows that, in any Bayes-Nash equilibrium, nobody converts away from the ruler's religion, while some convert into the ruler's religion if their religious fervour is low enough. In particular, $\overline{c} \in (0, 1)$ is such that

$$\overline{c} = p(\widehat{\mu}^*(\overline{c}))\eta_{r^*},$$

where

$$\widehat{\mu}^*(\overline{c}) = \begin{cases} p(\mu + (1-\mu)F(\overline{c})) \text{ if } r^* = m, \\ p(\mu(1 - F(\overline{c}))) \text{ if } r^* = \overline{m}. \end{cases}$$

□

**Remark 1.** *For general $p$ and $F$ functions, there could be multiple equilibria. In the Appendix, we show that a sufficient condition for there to be a unique Bayes-Nash equilibrium is that $p$ and $F$ are both differentiable, and that $|p'(\widehat{\mu})F'(c))| \leq 1$ for all $(\widehat{\mu}, c) \in [0,1] \times [0,1]$. We assume this condition holds henceforth.*

Next, we state and prove the main comparative statics' results that provide a theoretical basis for the empirical analysis. Without loss of generality, let $p(\widehat{\mu}) \equiv \varepsilon + h(\widehat{\mu})$ and interpret $\varepsilon \in (0,1)$ as the set of factors that influence the likelihood of an attack that are independent of the share of Muslims in the community (e.g., distance to the sea).

**Proposition 2.** *Increases in $\varepsilon$ increase the measure of conversions.*

*Proof.* The measure of conversions depends on the equilibrium threshold $\overline{c}$, which, in turn, is determined by equation $\chi(\overline{c}, \varepsilon) = 0$, where

$$\chi(\overline{c}, \varepsilon) \equiv \overline{c} - (\varepsilon + h(\widehat{\mu}^*(\overline{c}))\eta_{r^*}.$$

The result follows because $\chi$ is strictly increasing in $\overline{c}$ and strictly decreasing in $\varepsilon$.[9] □

Proposition 2 establishes that in communities which, for reasons exogenous to this model, are more likely to be attacked, there are more conversions to the ruler's religion. This result leads directly to the following statistical observation.

**Corollary 1.** *The correlation between $\varepsilon$ and the end-of-the-period share of Muslims is positive if the ruler's religion is Muslim and negative if it is non-Muslim.*

The next proposition establishes a comparative statics' result with respect to the ruler's preference for those with the same religion.

**Proposition 3.** *Increases in $\eta_{r^*}$ increase the measure of conversions.*

*Proof.* Analogous to the proof of Proposition 2. □

[9] We prove that $\chi$ is strictly increasing in $\overline{c}$ in the Appendix.

The previous proposition may help us test the model in two ways. First, rulers who have a stronger preference to protect their own vs rulers with weaker preferences (e.g., Islamic rulers in West Africa vs Islamic rulers in East Africa), and, second, communities that protect themselves better vs weaker communities (because overall ability to defend against attacks means that $\eta$ is smaller).

# 4 Data

## 4.1 Historical data

We estimate our first proxy for the spread Islam in Africa after the end of the slave trade by digitising a 1924 map by William R. M. Roome (1924), depicted in Figure A1.[10] Specifically, we recover information about the total land area in which Islam was practiced in 1924 (in 1,000 square km) and combine it with data on the locations of historical tribal boundaries mapped by George Peter Murdock (1967). Within our sample, Islam was practiced in about 61% of a tribe's land area and its penetration exhibited considerable variation: 157 ethnic groups had their whole tribal area Islamised, in 107 tribes Islam was not practiced, and in the remaining 197 tribes Muslims coexisted with people of different faiths. The map produced by Roome (1924) has been praised for his accuracy by contemporaries (see for instance a review by E.W.S., 1925) and has been used by other researchers studying the influence of Christian missions (Nunn et al., 2014; Lowes and Montero, 2021). To the best of our knowledge, the data on the spread of Islam has not been digitised so far.

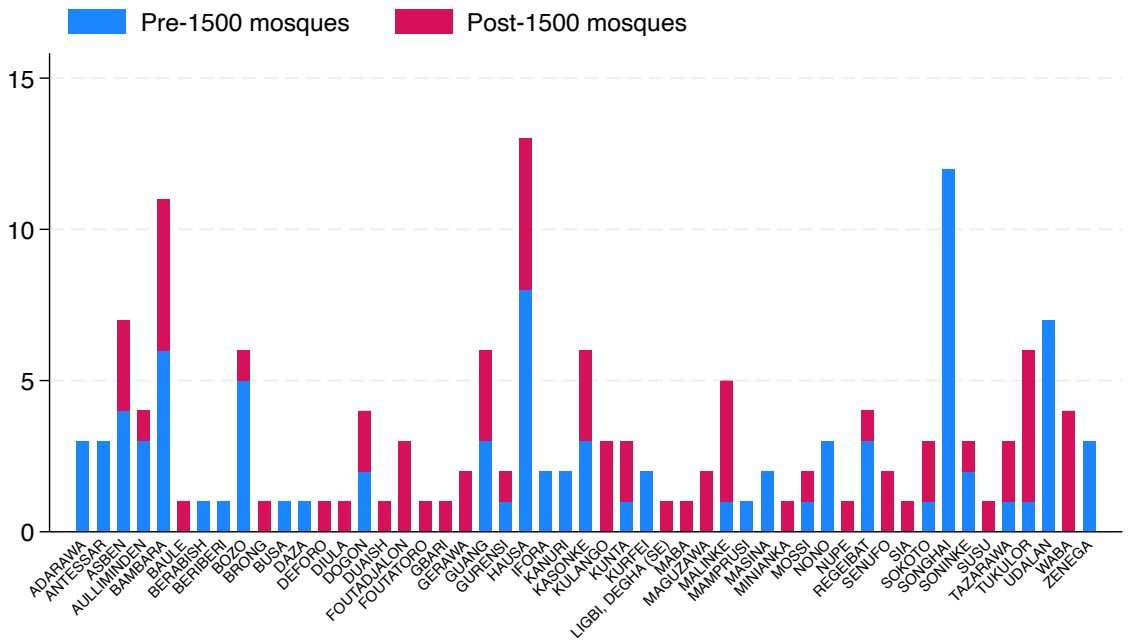
Next, we compute the strength of Islamic rulers in Africa after the Transatlantic slave trade using two proxies: the number of mosques established between the 16th and 19th century and an indicator capturing whether a tribe was ruled by an Islamic king during this period. The historical mosque data are from Pradines (2022), which provides information on the mosque name, its location and date of establishment, thus allowing us to identify Islamic presence before and during the Atlantic slave trade.[11] These mosques represent predominantly large congregational mosques, thus attesting not only the presence of a Muslim community, but also a symbol of political power by the local kings and sultans. We have geolocated 168 mosques in 51 tribal

[10] Roome was a surveyor and secretary of the British and Foreign Bible Society in East Africa. He crossed the continent several times and travelled 19,000 miles on foot and by bicycle (E.W.S., 1925).

[11] The information on the date of establishment often takes form of a range, thus making it unsuitable to use the data in panel format.

kingdoms in West Africa, corresponding to 11 modern countries (Burkina Faso, Chad, Ghana, Guinea, Ivory Coast, Mali, Mauritania, Niger, Nigeria, Senegal and Sierra Leone). 92 of them were established before the 16th century (see Figures 1 and 2): we use this group of mosques as additional control to measure the strength of pre-existing Islamic beliefs before the Atlantic slave trade.

Figure 1: Historical mosques among West African tribes, 7th-19th century




*Notes:* Number of mosques established before (blue) and after (red) the 16th century.
*Source*: Pradines (2022)

To construct the indicator of whether an ethnic group was ruled by an Islamic kingdom we digitsed three maps on the development of Muslim States in North and West Africa from Sluglett and Currie (2015). This source has been validated using Conrad (2010), Jedwab et al. (2022) and Iyigun (2007). Specifically, we have assigned a dummy equal to one if a tribal group's polygon intersected with that of an Islamic Kingdom.[12] To control for pre-existing adherence to Islam we also digitised kingdoms founded before the slave trade.[13] With our sample of 460 tribes, 223 of them were ruled by an Islamic king during 1500-1900, rising from 102 during the 1000-1500 period.

[12]The main Islamic kingdoms existing or established during 1500-1900 were: the Songhay of the Askiyas; Borno (Sayfi sultanate); Kanem; Tuareg sultanate of Agadez; Hausa; Futa Jala; Futa Toro; Sokoto Caliphate; Masina state; Tukolor empire.

[13]These were: Gao; Mali; Takrur kingdom; Kanem-Borno states; Songhai; Almoravids.

The data on the Atlantic slave trade are from Nunn (2008), and have been constructed by combining total slave export shipped from all ports of Africa with data about slaves' ethnicity. These figures have then been disaggregated by ethnic group, using the same procedure as Nunn (2008) and Nunn and Wantchekon (2011), which relies on Murdock (1967)'s classifications. The resulting estimates provide information on total Transatlantic slave exports at the tribe-level between the 16th and 19th centuries. Within our sample of West African tribes 9,295,498 slaves were exported to the Americas, thus representing around 74% of total estimated slaves export based on the Trans-Atlantic Slave Trade database.[14]

## 4.2 Modern data

To measure modern-day Muslim prevalence we use data from the Pew Forum on Religion and Public Life, conducted between December 2008 and April 2009, containing information on about 25,000 individuals from nineteen Sub-Saharan African countries, who were selected to reflect differences in colonial histories and religious compositions (Lugo and Cooperman, 2010). To construct the outcome variable, we assign an indicator equal to 1 if a respondent indicated Muslim as their preferred religion, and 0 if otherwise. In the survey, about 33.7% of respondents are Muslims, 62.3% are Christians and the remainder are traditionalist or do not belong to any religion.

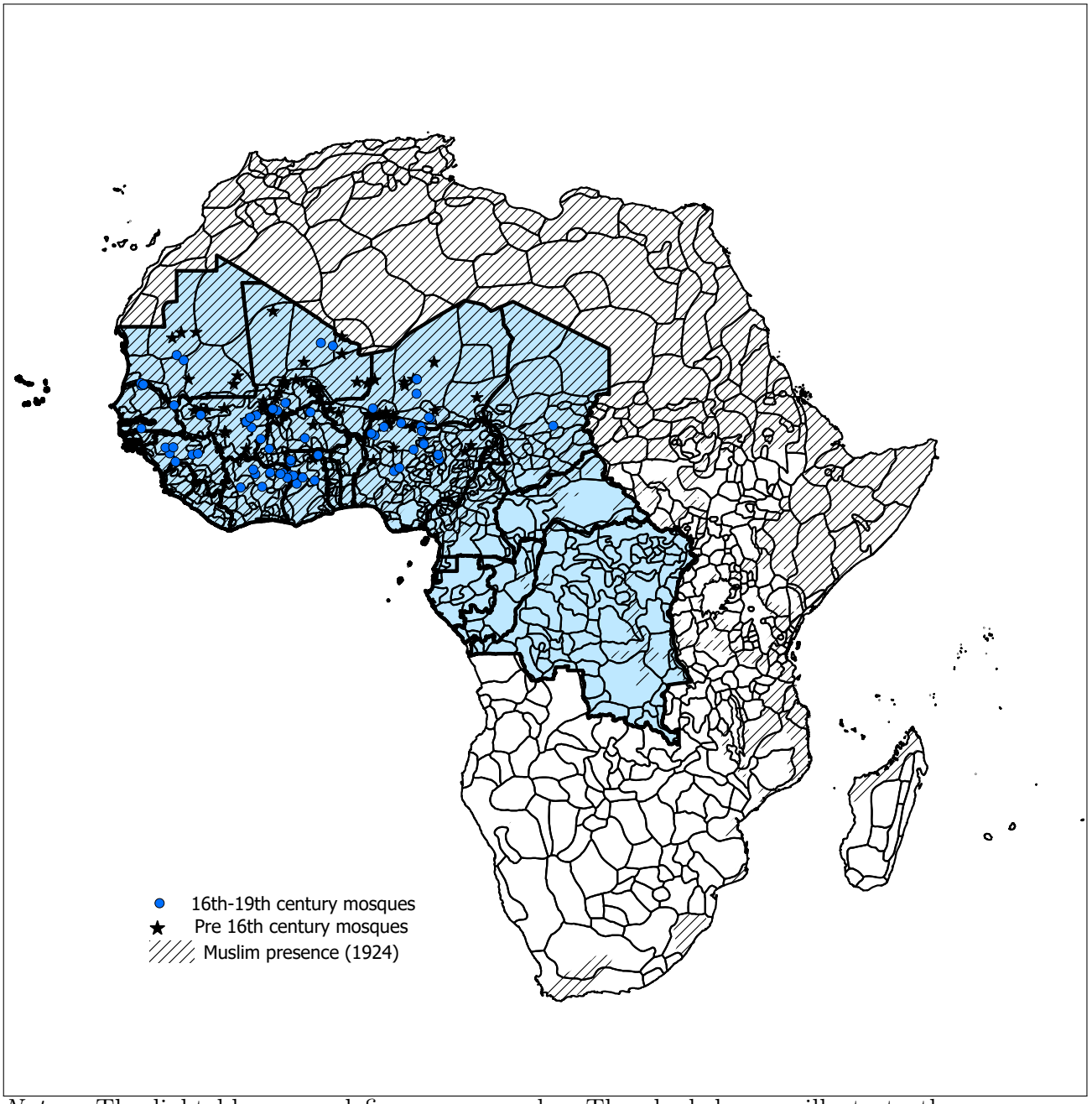
In addition to the religion indicator, the Pew survey contains information on respondents' education level, gender, region and ethnicity. The latter is instrumental for our analysis as it ensures a merge of the Pew data set with ethnic-level slave export data, our variable of interest, from Nunn and Wantchekon (2011). Similarly to the historical Islamic spread data, we match ethnic-level slave exports to respondents' ethnicity, resulting in about 300 unique matches.

Figure 3 illustrates the co-variation in Muslim prevalence and slave exports across ethnic groups in Sub-Sahara Africa; we distinguish between Atlantic slave exports (Figure 3a), involving predominantly West Africa and Indian slave exports (Figure 3b), largely concentrated in East and part of South-East Africa. The square grids present the shares of Muslim and the log of exports, while indicating the number of ethnic groups observed for each co-variation.

We observe significant variation between Muslim prevalence and slave exports across West, Central, East and Southern Africa. For example, in Figure 3a, nine ethnic groups (dark green, top row, right-most box) have close to 100% of all mem-

[14] Accessible at https://www.slavevoyages.org/voyage/database.

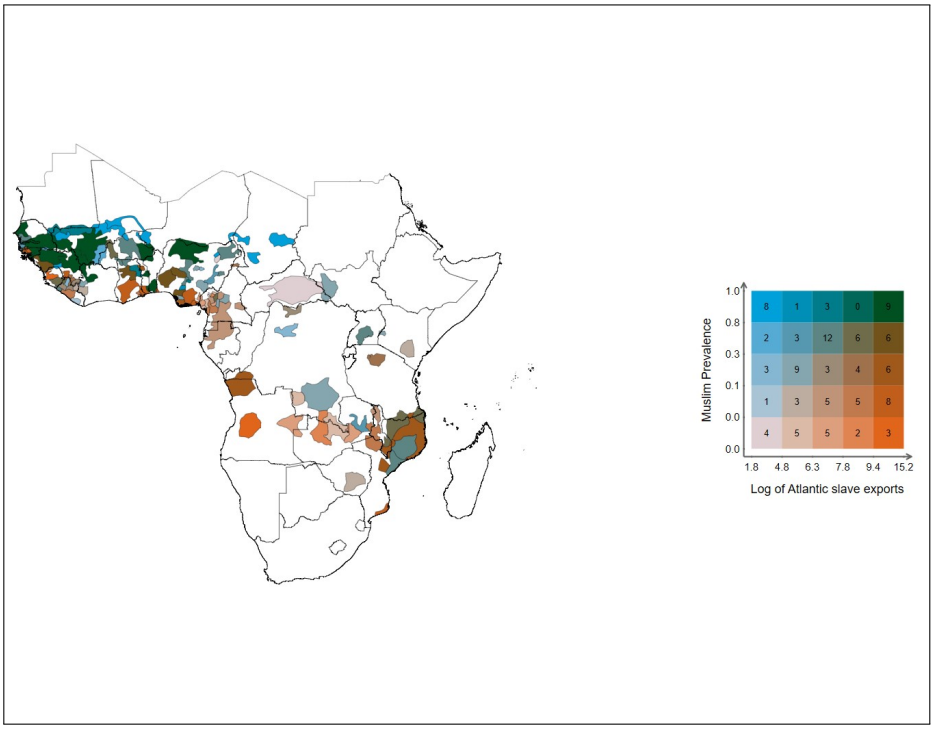
Figure 2: The spread of Islam in West Africa



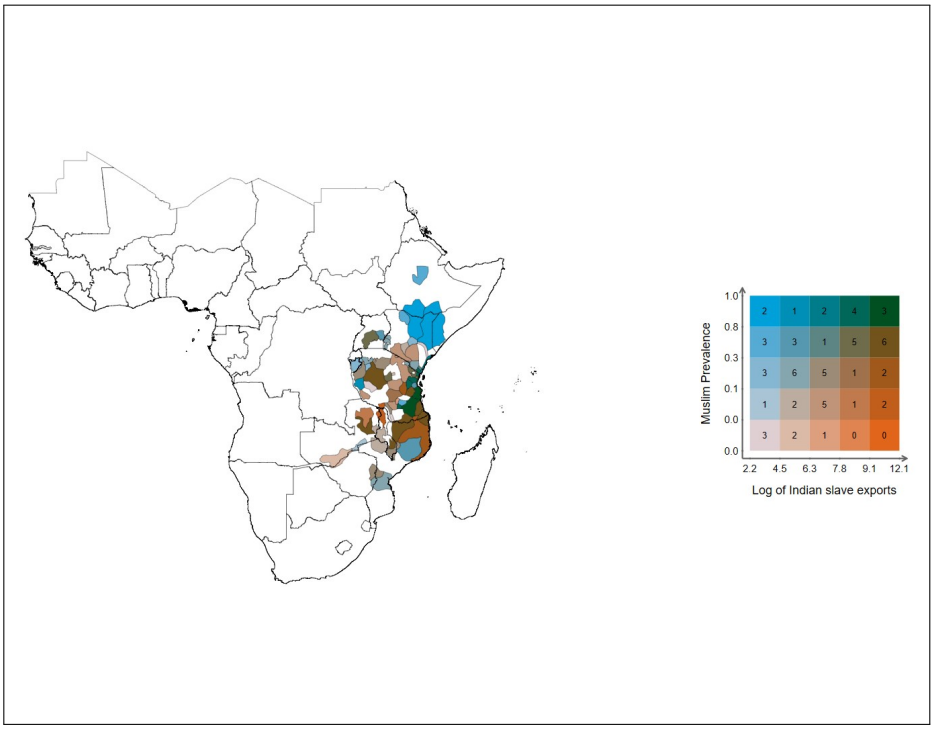

*Notes:* The light blue area defines our sample. The shaded areas illustrate the spread of Islam as of 1924, based on Roome (1924). The stars (circles) denote of mosques established before (after) the 16th century. *Source*: Pradines (2022)

bers being Muslim and about 4 million slaves ($e^{(15.2)}$) exported through the Atlantic Ocean. Meanwhile, about three ethnic groups (dark orange, second row from down, right-most box) have close to 10% of all members being Muslim and about similar number of slaves exported as well through the same channel. While these figures provide a visual understanding of the possible co-variation in our outcome variable

Figure 3: Co-variation between Muslim prevalence and slave exports




(a) Atlantic slave exports




(b) Indian slave exports

*Sources*: Slave exports from Nunn and Wantchekon (2011). Muslim prevalence from Lugo and Cooperman (2010).

and variable of interest, it may not be enough in establishing possible correlation between these variables.

# 5 Results

## 5.1 The spread of Islam after the 19th century

Table 1 presents the results on the relationship between Atlantic slavery and the geographical spread of Islam, proxied by the total land area in which Islam was practiced in 1924. We start with a correlation between the two variables (col.1), then we gradually add geographic controls (col. 2): latitude, longitude, an indicator if the tribe is on the coast, if it has a river, and the area of the tribal land in square km); two measures of pre-1500 Islamic spread (cols. 3-4): in Panel A a dummy variable equal to 1 if a tribe had a congregational mosque before 1500 (col. 3) and the number of such mosques (col.4) in Panel B a dummy variable equal to one if a tribe had an Islamic ruler before 1500 (col. 3) and the number of such rulers (col. 4); n. of years since a mosque had been established (col. 5); political controls (col. 6): an indicator for being engaged in local slavery and a measure of state centralisation; economic controls (col. 7): an indicator for practicing intensive agriculture, for plow use and adoption, for using large domesticated animals, for female playing a role in agriculture; measures for dependence on: gathering, hunting fishing, husbandry, agriculture, an agricultural suitability index.

The coefficient on Atlantic slave exports in Column (1) is negative and statistically significant, consistent with the historical narrative that fewer slaves were exported from Muslim regions. Moreover, the interaction terms between slavery and the strength of Islamic rulers – proxied by the presence of large congregational mosques (Panel A) and Islamic kingdoms (Panel B) – are positive and statistically significant across all specifications. These results underscore the protective role of Islamic rulers, which facilitated the diffusion of Islam in regions under their influence.

We assess the robustness of our estimates in Table 2. In Columns (1-2), we interact Atlantic slave exports with two measures of Islamic power: the number of mosques established post-1500 and the number of Islamic rulers in power post-1500. In Columns (3-4), we expand our analysis to account for the Indian slave trade, with the dependent variable and interactions calculated as the sum of both Atlantic and Indian slave exports. Finally, in Column (5), we test a triple interaction between Atlantic slave exports, the presence of post-1500 mosques, and Islamic rulers. The results remain consistent with our baseline estimates

Table 1: Atlantic slave trade and the spread of Islam

| | Dependent variable: Islamised tribe area in 1924 | | | | | | |
|---|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) |
| Panel A : Post-1500 mosque presence | | | | | | | |
| Atlantic slaves | -0.017*** | -0.018*** | -0.018*** | -0.018*** | -0.013*** | -0.010 | -0.010 |
| | (0.004) | (0.003) | (0.003) | (0.003) | (0.004) | (0.006) | (0.006) |
| Atlantic slaves × | | 0.045** | 0.040** | 0.043** | 0.070** | 0.112** | 0.119** |
| mosque presence | | (0.020) | (0.018) | (0.017) | (0.029) | (0.051) | (0.059) |
| Panel B : Islamic kingdoms | | | | | | | |
| Atlantic slaves | | -0.018*** | -0.018*** | -0.018*** | -0.014*** | -0.012** | -0.012** |
| | | (0.003) | (0.003) | (0.003) | (0.004) | (0.005) | (0.005) |
| Atlantic slaves × | | 0.022 | 0.023* | 0.023* | 0.030* | 0.053* | 0.058* |
| Islamic Kingdoms | | (0.014) | (0.014) | (0.014) | (0.016) | (0.028) | (0.030) |
| Geographic controls | N | Y | Y | Y | Y | Y | Y |
| Pre-1500 Islam Dummy | N | N | Y | Y | Y | Y | Y |
| Pre-1500 Islam Intensity | N | N | N | Y | N | N | N |
| Political controls | N | N | N | N | Y | Y | Y |
| Economic controls | N | N | N | N | N | Y | Y |
| *N* | 460 | 460 | 460 | 460 | 222 | 143 | 135 |

*Notes:*The dependent variable, measures the number of Atlantic slaves in thousands. Post-1500 mosque presence is a dummy variable equal to 1 if a tribe had a congregational mosque after 1500. Islamic Kingdom is a dummy variable equal to 1 if a tribe had an Islamic ruler. Pre-1500 Islam (dummy) represents a dummy variable equal to one if a mosque was established before 1500 (Panel A) and a dummy equal to 1 if a tribe had an Islamic ruler before 1500. Pre-1500 Islam (intesity) denotes the average number of years since when pre-1500 mosques have been established (Panel A) and the number of pre-1500 Islamic rulers. The geographic controls include: latitude, longitude, an indicator if the tribe is on the coast, an indicator if it has a river, and the area of the tribal land in square km. The political controls include: being engaged in local slavery, and a measure of state centralisation. The economic controls include: indicator for practicing intensive agriculture, for plow use and adoption, for using large domesticated animals, for female playing a role in agriculture; measures for dependence on: gathering, hunting fishing, husbandry, agriculture, an agricultural suitability index. All regressions are at the tribe level.

## 5.2 The spread of Islam: persistence

Table 2: Robustness to alternative specifications

| Dependent variable: | Islamised tribe area in 1924 | | | | |
|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) |
| Atlantic slaves | 0.016 | -0.018*** | | | -0.018*** |
| | (0.016) | (0.003) | | | (0.003) |
| Atlantic slaves × | 0.005** | | | | |
| n. post-1500 mosques | (0.002) | | | | |
| Atlantic slaves × | | 0.003* | | | |
| n. Islamic Kingdoms | | (0.002) | | | |
| Atlantic & Indian slaves | | | -0.018*** | -0.018*** | |
| | | | (0.003) | (0.003) | |
| Atlantic & Indian slave × | | | 0.041** | | |
| mosque presence | | | (0.017) | | |
| Atlantic & Indian slaves × | | | | 0.023* | |
| Islamic Kingdom | | | | (0.014) | |
| Atlantic slaves × mosque | | | | | 0.041** |
| presence × Islamic Kingdom | | | | | (0.017) |
| Geographic controls | Y | Y | Y | Y | Y |
| Pre-1500 Islam | Y | Y | Y | Y | Y |
| *N* | 460 | 460 | 460 | 460 | 460 |

*Notes:*The dependent variable in cols. (1), (4), (5) measures the number of Atlantic slaves in thousands, while in cols. (2)-(3) the number of Atlantic and Indian slaves in thousands. N. post-1500 mosques and n. Islamic Kingdoms denote the number of mosques and kingdoms established between the 15th and 19th centuries. Mosque presence (Islamic Kingdom) is a dummy variable equal to 1 if a tribe had a congregational mosque (Islamic ruler) after 1500. Pre-1500 Islam includes: the number of mosques established before 1500 and the average number of years since when pre-1500 mosques have been established. The geographic controls include: latitude, longitude, an indicator if the tribe is on the coast, an indicator if it has a river, and the area of the tribal land in square km. All regressions are at the tribe level.

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# 6 Model Appendix: On the existence and uniqueness of a symmetric Bayes-Nash equilibrium

As the text describes, there is Bayes-Nash equilibrium if there is some $\overline{c} \in [0,1]$ such that

$$\widehat{\chi}(\overline{c}) = 0, \tag{1}$$

where

$$\widehat{\chi}(\overline{c}) = c - p(\widehat{\mu}^*(\overline{c}))(\overline{\eta}_{r^*} - \underline{\eta}_{r^*}).$$

That such a threshold $\overline{c}$ exists is ensured by the fact that $\widehat{\chi}(0) < 0$ and that $\widehat{\chi}(1) > 0$.

To discuss uniqueness, notice that

$$\widehat{\chi}'(\overline{c}) = \begin{cases} 1 - (1-\mu)p'(\widehat{\mu}^*)F'(\overline{c}) \text{ if } r^* = m, \\ 1 + p'(\widehat{\mu}^*)F'(\overline{c})\mu \text{ if } r^* = \overline{m}. \end{cases}$$

It then follows that if the condition of the text holds - $|p'(\widehat{\mu})F'(c))| \leq 1$ for all $(\widehat{\mu}, c) \in [0,1] \times [0,1]$ - then $\widehat{\chi}$ is strictly increasing, so that there is at most one solution to equation 1.